

On the Edge of Stability: Spiking Reservoir State-Space Encoding of Affective EEG

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Abstract—How a representation of affective event-related potentials (ERPs) degrades under perturbation can depend on the evaluation protocol as much as on the representation. We encode affective ERPs with a fixed leaky integrate-and-fire spiking reservoir operated near its measured edge of stability ($\rho^* \approx 0.82$, below the echo-state heuristic $\rho = 1$; removing the recurrence costs 0.049 balanced accuracy, BA, 95% CI $[-0.098, -0.002]$) and compare it with band-power, ERP-window, and trained EEGNet representations under subject-grouped validation with subject-level uncertainty. Two scoring effects distort channel-dropout comparisons: raw retention rewards encoders near the chance floor, and zero-filling a dropped channel before standardization is not invariant to re-centering the features, so at 30% dropout it halves the measured lead of ERP-window features over the near-centered reservoir (+0.05 versus +0.09 to +0.11 BA), and on DEAP the fill rule moves band-power BA by up to 0.11. When the raw signal itself is corrupted, identically for every encoder, ERP-window features stay ahead of the reservoir, whose fill invariance does not survive removal of half the electrodes, and channel-dropout augmentation more than halves EEGNet’s loss to electrode removal. We recommend reporting absolute accuracy with subject-level uncertainty, stating and varying the fill rule, and reporting trained models with and without augmentation.

Index Terms—EEG, event-related potentials, spiking neural networks, reservoir computing, affective computing, neuromorphic computing, ARSPI-Net.

I. INTRODUCTION

Electroencephalographic event-related potentials (ERPs) provide millisecond-scale measurements of evoked neural dynamics, but the recorded waveform is a noisy, low-dimensional, and subject-dependent projection of an unobserved neural process [14], [15]. A representation may separate affective conditions under one protocol while preserving little usable structure after amplitude noise, temporal misalignment, or channel loss.

Endpoint accuracy alone also says little about what a representation preserves, which matters for spiking encodings. Fixed recurrent reservoirs and spiking networks provide nonlinear temporal expansions [1], [4], but EEG/ERP studies often treat them primarily as classifiers [11], [17], and their spike-count features are rarely examined as perturbation-sensitive temporal measurements. How such encodings change under perturbation has received little attention, even though preprocessing, acquisition, and electrode-loss choices can materially change EEG representation performance [18]–[20]. Channel-dropout comparisons that zero-fill a dropped channel also do not account for the coordinate origin of the features, which, as we show, can change the measured result.

This paper evaluates whether channel-dropout robustness rankings for affective-ERP representations are stable under reasonable changes to the scoring protocol. We compare spectral band-power, ERP-window amplitudes, a fixed leaky integrate-and-fire (LIF) reservoir operated near the edge of its ordered regime and summarized by six spike-count bins (BSC_6), and a trained EEGNet. The reservoir is not treated as a proposed winner; it is used as a fixed reference whose embedding is nearly zero-centered and whose dynamics are not retrained under perturbation. The contributions are:

- 1) A subject-grouped comparison of affective-ERP representations under electrode removal, amplitude noise, and temporal jitter applied identically to every encoder’s input, with uncertainty and encoder contrasts from a subject-level bootstrap rather than from folds treated as independent (Secs. V-A, V-C; Tables I, III).
- 2) Quantification of two protocol artifacts in channel-dropout evaluation: raw-retention inflation near the chance floor, and zero-imputation bias for non-centered features, which is not invariant to re-centering (Proposition 1) and is also tested on DEAP (Secs. III-E, V-B, V-D).
- 3) Paired subject-level evidence that zero-fill halves the measured lead of ERP-window features over the near-centered reservoir, and that channel-dropout augmentation, a training choice, more than halves the trained EEGNet’s loss under electrode removal (Table II, Fig. 3, Sec. V-C).
- 4) A test of the reservoir’s operating point: a spectral-radius sweep with a no-recurrence ablation and a measured order parameter places the transition at $\rho^* \approx 0.82$, below the echo-state heuristic $\rho = 1$ and next to the pre-specified $\rho = 0.9$, inside a broad accuracy optimum (Fig. 1).

Affective categories define the task; the representations and their invariances under perturbation are the object of study.

II. RELATED WORK

ERP/EEG analysis commonly uses window amplitudes, Hjorth [12] and wavelet/spectral descriptors [13], and linear models [15], which are interpretable through known components such as the P300 and late positive potential [14] but can miss nonlinear temporal structure. Reservoir computing offers a complementary view: a fixed high-dimensional recurrent system maps the input to a state trajectory read out by a simple

classifier [1]–[4], and spiking liquid-state machines add event-based timing and count structure [7], [8]. Reservoir computation is often reported to peak near the transition between ordered and chaotic dynamics, located by how far a small state perturbation spreads [5], [6]; for a thresholded spiking reservoir that transition need not coincide with $\rho = 1$, the echo-state heuristic for rate reservoirs. With fixed dynamics, the reservoir acts as a measurement operator rather than a classifier.

Spiking and deep networks are now widely applied to EEG [11], [16], [17]. Much of this literature emphasizes endpoint accuracy over preprocessing and acquisition choices [20], [21]. Neuromorphic reservoir features for affective EEG were introduced in prior ARSPI-Net work [9], [10]; here we establish their perturbation and invariance properties and identify two scoring artifacts that distort robustness comparisons.

Dynamic spatial filtering handles missing channels with a learned attention front end [23], and channel-dropout or amplitude-noise augmentation hardens trained models [24]. A recent EEG foundation-model benchmark finds that apparent channel-dropout fragility can depend on whether dropped channels are removed or zero-padded [22]. More broadly, missing-data handling in neuroimaging spans a spectrum from constant fills and spherical-spline interpolation of absent electrodes [25], through learned imputation and learned spatial filtering [23], to conditional generative synthesis of entire missing modalities with ViT-GAN and latent-diffusion models [26], [27]. The four fills we study sit at the simple, non-learned end of that spectrum. We separate two scoring effects, raw-retention inflation and zero-fill bias, and use the training-free, near-centered reservoir as a reference.

III. METHODS

A. ERP Observation Model

Let $X_s^{(c)} \in \mathbb{R}^{C \times T}$ denote the trial-averaged ERP observation for subject s and condition c ($C = 34$ channels, $T = 256$ samples). We treat it as a noisy readout of an unobserved neural state trajectory,

$$\xi_{t+1} = f_\theta(\xi_t, u_t) + \eta_t, \quad x_t = H\xi_t + \epsilon_t, \quad (1)$$

where H is the scalp measurement operator and ϵ_t is measurement and preprocessing noise. The perturbations below act on this model (Sec. V-B).

B. Spiking Reservoir Encoding

For each electrode trace, the signal drives a fixed LIF reservoir of 256 neurons with input weights W_{in} , recurrent weights W , leak $\beta = 0.05$, and threshold $\vartheta = 0.5$:

$$v_t = [(1 - \beta)(1 - z_{t-1}) \odot v_{t-1} + Wz_{t-1} + W_{\text{in}}x_t]_+, \quad (2)$$

$$z_t = \mathbf{1}[v_t \geq \vartheta], \quad (3)$$

where v_t is the membrane potential, z_t the binary spike vector, and $[\cdot]_+$ rectification at zero; the factor $(1 - z_{t-1})$ resets every unit that spiked at the previous step (the implementation also subtracts ϑ from a unit that has just spiked, which changes

no spike). W_{in} and W are drawn once from a Xavier-uniform law (seed 42), and W is rescaled to a pre-specified spectral radius $\rho = 0.9$; only the readout is trained. To test whether the recurrent dynamics matter, the same reservoir is rescaled to $\rho \in \{0, 0.3, 0.6, 0.9, 1.2, 1.5\}$ with every other parameter fixed ($\rho = 0$ removes the recurrence, leaving a feedforward LIF layer). A label-free order parameter locates the dynamical transition by damage spreading [5]: two copies receive the same electrode drive, one unit’s spike is flipped at the start of the analysis window, and the normalized Hamming distance between the copies’ spike vectors, averaged over the steps more than ten after the flip to the end of the window and over 400 sampled drives and flipped units, shows whether the flip dies out (ordered) or spreads (irregular). The transition ρ^* is the ρ at which the damage, linearly interpolated between grid points, reaches half its maximum over $\rho \in [0, 2]$.

Activity is summarized by six equal temporal bins over the window $t \in [10, 70]$ in sample units; with stimulus onset at sample 51 (Sec. IV-A), this spans about -160 to $+73$ ms around onset. For reservoir unit j and bin k , the binned spike-count code is $b_{j,k} = \sum_{t \in \mathcal{B}_k} z_j(t)$, $k = 1, \dots, 6$. Each electrode’s spike-bin vector is projected onto a single 64-component PCA basis shared across electrodes, and the 34 projections are concatenated into a trial-level representation $E_s^{(c)} \in \mathbb{R}^{2176}$. The encoding is a composition of a fixed non-linear temporal map, spike-count binning, and an unsupervised projection:

$$E = \phi_{\text{PCA}}([b_1, \dots, b_6]) = \Phi(X), \quad (4)$$

a label-free representation whose PCA basis is fitted on the training subjects of each fold only; fitting it on all subjects (transductive) is reported as a sensitivity check (Sec. V-A). A single-bin control (BSC₁, total spike count in the window) uses the identical pipeline. The window lies mostly before onset and excludes the later components the ERP-window baseline sees. Because every epoch is z-scored over its full length, pre-stimulus samples also carry an offset set by the post-stimulus signal, so we do not attribute the reservoir’s accuracy to a specific evoked component.

C. Baselines and Readout

Two classical baselines are evaluated under the same subject-grouped protocol. The band-power baseline (A0) uses Welch band power in five bands (1–4, 4–8, 8–13, 13–30, 30–45 Hz) per channel ($5 \times 34 = 170$ dimensions). The ERP-window baseline (A0_e) uses mean amplitude in three fixed windows, about -120 – 0 ms (pre-stimulus baseline), 50 – 250 ms (early components), and 250 – 600 ms (P300 and late positive potential) relative to onset, per channel ($3 \times 34 = 102$ dimensions). The band-power, ERP-window, and reservoir representations use a balanced L2-regularized logistic readout to test separability without a high-capacity classifier.

As a trained deep baseline we evaluate EEGNet-8,2 [16] on the raw 34×256 ERP signal, trained per fold (Adam, 80 epochs) with train-only per-channel standardization, with and

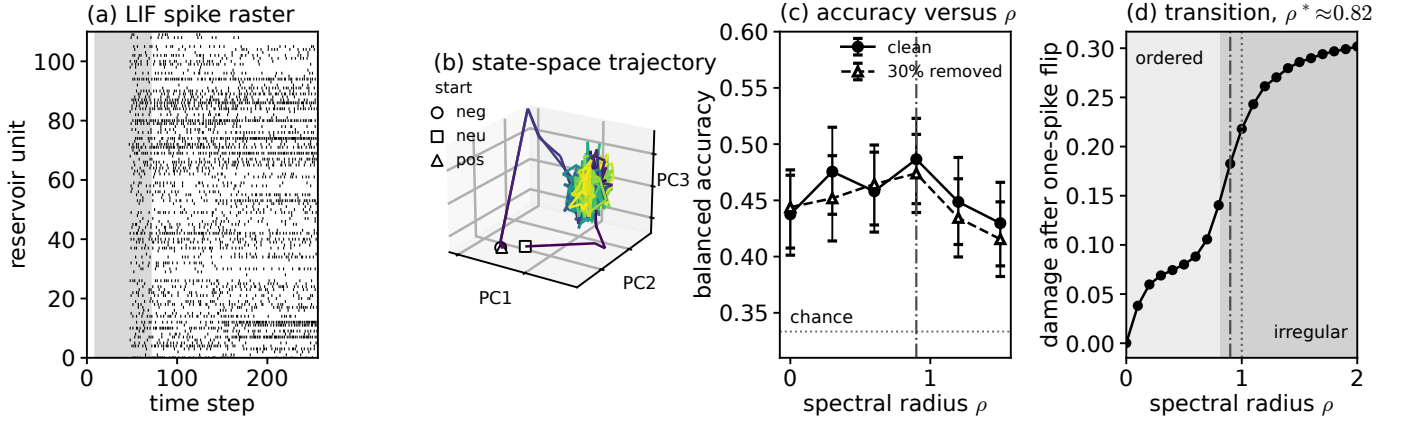


Fig. 1. The fixed LIF reservoir as a dynamical system, linked to accuracy. (a) Spike raster (units 0–109 of 256) for one condition (shaded = the $t \in [10, 70]$ analysis window, about -160 to $+73$ ms around onset at step 51). (b) Population state-space trajectory (first three principal components of the 256-unit membrane state), colored from epoch start (dark) to end (light), for the three affective conditions; the trajectories largely overlap, consistent with the modest clean separability. (c) Subject-level BA (95% CI) versus spectral radius ρ , clean and with 30% of electrodes removed; $\rho = 0$ removes the recurrence, and the dash-dot line marks the pre-specified $\rho = 0.9$. (d) Damage-spreading order parameter versus ρ ; shading separates the ordered and irregular regimes at the measured transition $\rho^* \approx 0.82$; dash-dot = $\rho = 0.9$, dotted = the echo-state heuristic $\rho = 1$.

without augmentation by random channel dropout and additive noise.

D. Perturbation Diagnostics

We separate two experiments. The feature-coordinate experiment (Sec. V-B) suppresses a dropped channel’s feature block after extraction and refills it by one of four rules; it isolates the fill convention and the coordinate origin, and applies to the fixed encoders. The signal-level experiment (Sec. V-C) corrupts the test subjects’ raw signal before any feature extraction, identically for every encoder including EEGNet with the same draws in each fold, while training uses the clean signal: removal of 10, 30, or 50% of the electrodes, additive white noise at 5 dB SNR per channel, and a uniform alignment shift of up to ± 50 ms, with all features recomputed from the corrupted signal. In terms of Eq. (1), noise augments ϵ_t , removal zeroes rows of H , and jitter shifts the sampling index of x_t . Performance is balanced accuracy (BA); Table I also reports macro-F1 and macro one-vs-rest ROC-AUC.

E. Retention Scores and Two Confounds

Raw retention, the ratio of perturbed to clean BA, can overstate the stability of representations whose clean BA is close to chance ($c = 1/3$ for three balanced classes), so we also report *above-chance retention*, $(\text{BA}_{\text{pert}} - c)/(\text{BA}_{\text{clean}} - c)$.

Channel dropout also requires a fill rule for the dropped channel’s features. We compare the common zero-fill convention with train-mean imputation, then add two in-distribution checks: train-fitted k -nearest-neighbor imputation ($k=10$) and a within-sample spatial fill using the mean of retained channels.

Non-invariance of zero-fill robustness. Write the standardized readout input as $z = D_\sigma^{-1}(x - \mu)$ with $D_\sigma = \text{diag}(\sigma)$, and let S index the dropped coordinates. Zero-fill sets $x_S = 0$, so the dropped block enters as $z_S = -D_{\sigma,S}^{-1}\mu_S$; train-mean fill

sets $x_S = \mu_S$, so $z_S = 0$. The two fills therefore differ by a fixed shift,

$$z_S^{\text{zero}} - z_S^{\text{mean}} = -D_{\sigma,S}^{-1}\mu_S. \quad (5)$$

Proposition 1. *Let the clean readout depend on x only through z , so it is invariant to a constant offset $x \mapsto x - b$ (train-only standardization guarantees this). Then the clean prediction is unchanged by the re-centering, while the zero-fill dropout input shifts by $\Delta z_S = D_{\sigma,S}^{-1}b_S$. Hence the measured zero-fill dropout accuracy is not a function of the information in x alone; it depends on the coordinate origin, and the zero- and mean-fill inputs coincide if and only if $\mu_S = 0$.*

Proof. Standardization subtracts the training mean, so b cancels from the clean input. A zero-filled coordinate maps to $z_i = -(\mu_i - b_i)/\sigma_i$ instead of $-\mu_i/\sigma_i$, a shift of $b_i/\sigma_i \neq 0$ for any $b_i \neq 0$, and zero- and mean-fill coincide exactly when $\mu_S = 0$. $\square$

For a linear readout with weights w , Eq. (5) changes the dropped-block contribution to the logit by $\sum_{i \in S} w_i \mu_i / \sigma_i$ between the two fills, at most $\|w_S\| \|D_{\sigma,S}^{-1}\mu_S\|$ in magnitude, so the fills can disagree only as far as the dropped features are non-centered, and only a centered code makes the choice immaterial. The reservoir’s PCA code is near-centered: averaged over features and training folds, $|\mu_i|/\sigma_i$ is 0.06 for the reservoir, against 0.75 for ERP-window and 1.20 for band-power, which is why it serves as a reference in Sec. V-B.

IV. EXPERIMENTAL PROTOCOL

A. Dataset and Task

The experiments use trial-averaged affective ERP data from the Stress, Health, and the Psychophysiology of Emotion (SHAPE) project [29], collected by the Laboratory for Clinical Affective Neuroscience at Stony Brook University, with 211 subjects after exclusion of one anomalous subject record. Each subject contributes three affective conditions (negative, neutral, pleasant), yielding 633 subject-condition observations. Each

TABLE I

CLEAN THREE-CLASS BALANCED ACCURACY FOR THE FIXED ENCODERS, WITH SUBJECT-LEVEL BOOTSTRAP 95% CIs (211 SUBJECTS RESAMPLED WITH REPLACEMENT, THREE CONDITIONS EACH), MACRO-F1, AND MACRO ONE-VS-REST ROC-AUC FROM THE POOLED OUT-OF-FOLD PROBABILITIES. CHANCE IS 1/3 (AUC 0.5). EEGNet IS IN TABLE III.

Cfg	Features	BA [95% CI]	F1	AUC
A0 _e	ERP-window (102)	0.629 [0.597, 0.660]	0.629	0.799
A0	Band-power (170)	0.494 [0.460, 0.528]	0.492	0.682
A1	Reservoir BSC ₆ (2176)	0.487 [0.447, 0.523]	0.485	0.658

recorded epoch (34 scalp channels at 1024 Hz, 1.2 s, baseline-corrected over the 200 ms pre-stimulus interval) is analyzed as every fourth of its first 1024 samples: 256 samples at 256 Hz spanning -200 to $+800$ ms around stimulus onset (onset at sample 51), z-scored per channel within each epoch; the grand averages show the expected late-positive-potential enhancement for emotional over neutral stimuli (Fig. 2). Subject-level data are maintained in a restricted research environment; only aggregate, de-identified results are reported.

B. Validation

All results share one fold structure, StratifiedGroupKFold (5 folds, seeds 42–46) with subjects as groups, so observations from the same subject never span train and test; standardization is fitted on the training subjects, the fixed encoders use a balanced L2 logistic readout, and EEGNet is trained on the same folds. Uncertainty is estimated at the level of the subject, not the fold. For each representation we average the out-of-fold class probabilities of every observation across the five repeated partitions, then form 95% confidence intervals (CIs) from 4000 resamples of the 211 subjects with replacement, keeping all three condition observations of a resampled subject together. Comparisons between encoders, fills, or conditions are paired differences under the same subject resampling. Because the five partitions are pooled into one prediction per observation before resampling, the dependence among the 25 overlapping fold estimates, which reuse the same subjects, does not enter the intervals, and no test treats folds as independent samples. Chance is 1/3, and a within-subject label-permutation null (20 permutations) gives mean BA 0.33 to 0.34 (SD at most 0.023, maximum 0.39) for every fixed encoder.

V. RESULTS

A. Clean Classification and Controls

Table I reports clean three-class BA, macro-F1, and macro one-vs-rest ROC-AUC for the fixed encoders. ERP-window features are the strongest fixed encoder (BA 0.629), above band-power (0.494) and the reservoir embedding (0.487), whose subject-level intervals overlap. EEGNet, trained on the raw signal under the same folds, is more accurate still (BA 0.731, [0.701, 0.763]; Table III). The fixed encoders therefore carry genuine but modest condition information, and the reservoir is not the most accurate encoder.

Spectral radius and recurrent dynamics. The recurrent dynamics contribute: relative to the pre-specified $\rho = 0.9$, the feedforward layer ($\rho = 0$) loses 0.049 BA (95% CI $[-0.098, -0.002]$), and the strongly irregular $\rho = 1.5$ loses 0.057 ($[-0.106, -0.011]$; -0.058 , $[-0.103, -0.016]$, with 30% of electrodes removed; Fig. 1c). Between $\rho = 0.3$ and 1.2 the differences from $\rho = 0.9$ are within noise, as is the $\rho = 0$ contrast with 30% of electrodes removed (-0.030 , $[-0.074, +0.016]$), so the optimum is broad rather than sharp. After an initial step at small ρ , the order parameter rises steadily, with its largest increment between $\rho = 0.8$ and 0.9 and saturation above $\rho \approx 1.2$; its half-maximum crossing places the measured transition at $\rho^* \approx 0.82$, below the echo-state heuristic $\rho = 1$ (Fig. 1d); the mean firing rate doubles across it (0.06 at $\rho = 0.6$, 0.12 at $\rho = 0.9$). The pre-specified operating point thus lies just past the measured edge of the ordered regime, inside a broad accuracy optimum.

Temporal-binning ablation. A single-bin total-count control (BSC₁) matches BSC₆: the paired difference BSC₆ minus BSC₁ is +0.001 BA (95% CI $[-0.044, +0.047]$) on clean data and -0.001 ($[-0.046, +0.041]$) with 30% of electrodes removed. The reservoir’s accuracy comes from the fixed state-space expansion, not from six-bin temporal resolution, and we do not claim a temporal-coding benefit for BSC₆.

Transduction and component controls. Fitting the PCA basis on all subjects (transductive) instead of the training subjects gives BA 0.485 [0.447, 0.521], against 0.487, and 16, 32, 64, or 128 components per electrode give 0.477 to 0.507, all inside the 64-component interval [0.447, 0.523]. Neither the basis nor the component count accounts for the results.

B. Feature-Coordinate Experiment: Two Scoring Confounds

Channel dropout is where the reported conclusion depends on the scoring convention (Table II, Fig. 3).

Accuracy-floor effect. Raw retention makes the reservoir look the most robust encoder: at 30% dropout its BA does not move (retention 1.00, against 0.85 for ERP-window under zero-fill). This reflects the ratio, not stability: the reservoir sits close to the 1/3 chance floor and has little above-chance accuracy to lose. Above-chance retention corrects the floor, but its denominator is only about 0.15 for the reservoir, so it is unstable near chance and can exceed one. We therefore treat absolute BA and paired subject-level differences as primary, with retention ratios as secondary diagnostics.

Zero-imputation effect. By Proposition 1 the two constant fills can disagree only through the non-centeredness of the dropped features, and the subject-level gaps are ordered as that non-centeredness is (Fig. 3a). At 30% dropout, mean-fill exceeds zero-fill by +0.055 BA (95% CI $[+0.024, +0.087]$) for band-power and by +0.042 ($[+0.011, +0.074]$) for ERP-window, but by -0.000 ($[-0.022, +0.022]$) for the near-centered reservoir. Across rates from 10 to 50% the reservoir’s two constant-fill scores agree within 0.016 BA and no CI excludes zero, whereas the gap of the non-centered encoders varies with the rate and is significant for band-power at 20 and 30% and for ERP-window at 30%. k NN fill differs in kind

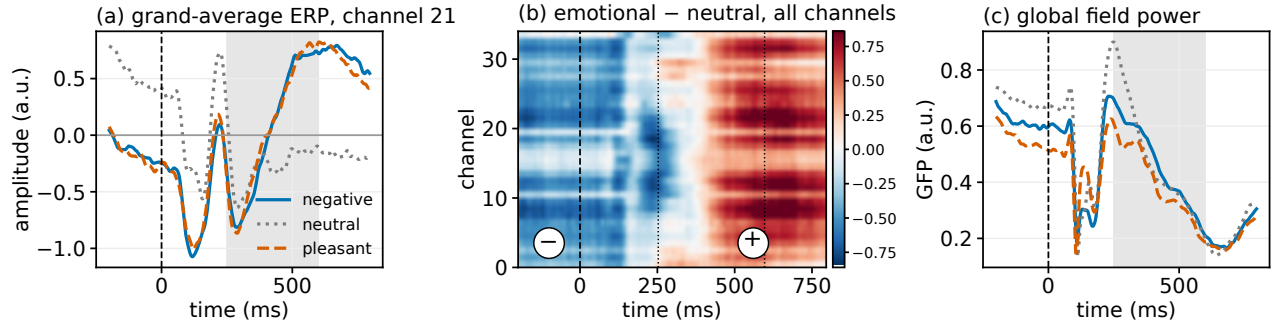


Fig. 2. Trial-averaged affective ERP data (SHAPE cohort; grand averages over 211 subjects). (a) Grand-average waveforms per condition at the most affect-discriminative channel; time 0 = stimulus onset (dashed), shaded = the late ERP-window feature (250–600 ms). (b) Spatiotemporal map of the emotional-minus-neutral difference across all 34 channels (dotted = the same window; -/+ mark the sign), showing the sustained LPP enhancement. (c) Global field power per condition.

TABLE II

CHANNEL DROPOUT AT 30% (FEATURE-COORDINATE EXPERIMENT): SUBJECT-LEVEL BALANCED ACCURACY FOR THE FIXED ENCODERS UNDER FOUR FILL RULES, AND THE PAIRED DIFFERENCE ERP-WINDOW MINUS RESERVOIR UNDER THE SAME SUBJECT RESAMPLING. A POSITIVE DIFFERENCE WHOSE 95% CI EXCLUDES ZERO MEANS ERP-WINDOW IS SIGNIFICANTLY AHEAD. PER-ENCODER 95% CIs ARE ± 0.03 TO ± 0.04 .

Fill	Band-pow.	ERP-win.	Reserv.	ERP-Res [95% CI]
Zero	0.417	0.536	0.487	+0.049 [+0.003, +0.095]
Mean	0.472	0.578	0.487	+0.092 [+0.049, +0.134]
kNN	0.483	0.610	0.507	+0.103 [+0.062, +0.147]
Spatial	0.488	0.589	0.482	+0.107 [+0.063, +0.150]

(Fig. 3b): it replaces the dropped block by an average over ten training neighbors, which adds content rather than moving the origin, and at 40 and 50% it significantly raises every encoder, ERP-window most (+0.13 BA at 50%, [+0.09, +0.16]).

Encoder comparison. ERP-window features lead the reservoir under every fill, but the fill sets the size of the lead (Table II). At 30% dropout the paired difference ERP-window minus reservoir is +0.049 BA under zero-fill (95% CI [+0.003, +0.095]) and +0.092 to +0.107 under the three in-distribution fills, whose CIs all lie above +0.04. Zero-fill thus halves the measured advantage of the strongest fixed encoder, because only the non-centered encoder’s score moves with the constant.

C. Signal-Level Perturbations

The signal-level experiment asks what happens when the recorded signal itself is corrupted (Table III). A removed electrode contributes zero band-power and zero ERP amplitude, so for these two encoders removal is exactly zero-fill, and their removal rows equal the zero-fill scores of Sec. V-B. For the reservoir, a silent electrode yields a zero spike-count block, which projects to the negative of the projected training mean rather than to the PCA origin, so the shift of Eq. (5) reappears before the projection.

Three results stand out. First, ERP-window features remain the most accurate fixed encoder under every condition; their paired lead over the reservoir ranges from +0.06 to

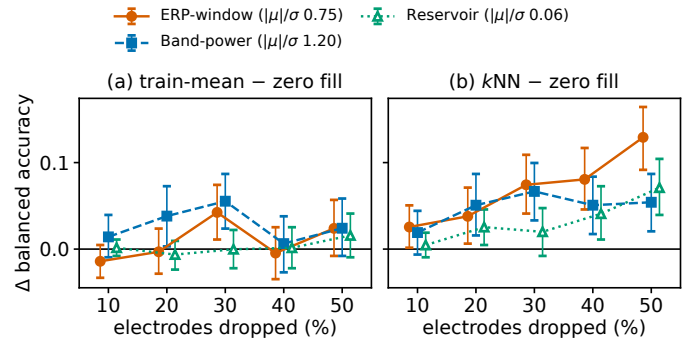


Fig. 3. How much the fill rule alone moves channel-dropout accuracy (feature-coordinate experiment): paired subject-level change in BA relative to zero-fill, with 95% CIs, versus the share of electrodes dropped. (a) Train-mean fill, which differs from zero-fill only by the shift of Eq. (5); the legend gives the mean $|\mu|/\sigma$ of each encoder’s features. (b) kNN fill, which replaces the dropped block by an average of ten training neighbors and so changes its content.

+0.14 BA, and every CI excludes zero. Second, the reservoir holds its accuracy through 30% removal (loss 0.013, 95% CI [−0.014, +0.041]) but falls to 0.393 at 50%, close to chance (loss 0.093, [+0.055, +0.131]), so its fill invariance is a property of its feature space, not robustness to missing electrodes. Third, 5 dB noise costs band-power 0.059 BA ([+0.019, +0.098]) while ERP-window and the reservoir stay within noise of their clean scores, and ± 50 ms jitter moves no fixed encoder, consistent with the wide ERP windows and with the single-bin control of Sec. V-A.

Training effect. EEGNet ingests the same corrupted signal and is the most accurate encoder in every condition (clean BA 0.731, 95% CI [0.701, 0.763]; paired lead over ERP-window +0.08 to +0.20, every CI excluding zero). Unaugmented, it loses 0.071 BA at 30% removal and 0.144 at 50%, close to the ERP-window losses (0.093 and 0.155). Training with channel-dropout and noise augmentation leaves clean accuracy unchanged (paired difference −0.000, [−0.025, +0.022]) and cuts these losses by more than half, to 0.028 and 0.062: the augmented network is ahead by +0.043 ([+0.009, +0.077]) at 30% and +0.082 ([+0.049, +0.115]) at 50%, while neither

TABLE III

SIGNAL-LEVEL PERTURBATIONS: SUBJECT-LEVEL BALANCED ACCURACY WITH EVERY REPRESENTATION RECOMPUTED FROM THE CORRUPTED TEST SIGNAL (SAME DRAWS FOR ALL ENCODERS; TRAINING ON CLEAN DATA, EEGNet ALSO WITH AUGMENTATION). PER-CELL 95% CIs ARE ± 0.03 TO ± 0.04 . CHANCE IS 1/3.

Condition	Band-pow.	ERP-win.	Reserv.	EEGNet	+aug.
Clean	0.494	0.629	0.487	0.731	0.731
10% removed	0.476	0.602	0.469	0.719	0.722
30% removed	0.417	0.536	0.474	0.660	0.703
50% removed	0.450	0.474	0.393	0.588	0.670
Noise 5 dB	0.436	0.619	0.479	0.738	0.727
Jitter ± 50 ms	0.496	0.626	0.493	0.719	0.708

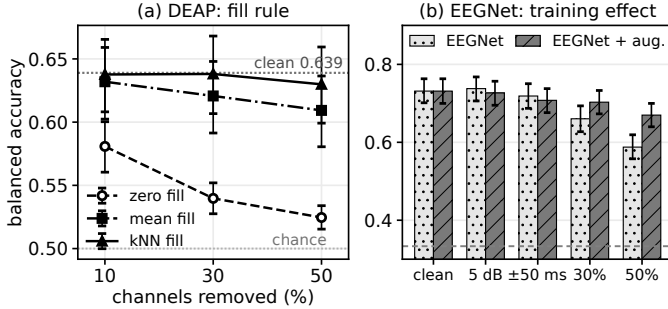


Fig. 4. (a) External check on DEAP [28] (32 subjects, within-subject binary valence, band-power; clean BA 0.639; CIs from resampling subjects). Under channel removal, BA depends strongly on the fill rule. (b) EEGNet without and with channel-dropout and noise augmentation under the signal-level conditions of Table III (clean, 5 dB noise, ± 50 ms jitter, 30% and 50% of electrodes removed; subject-level 95% CIs). Augmentation leaves clean accuracy unchanged, moves noise and jitter accuracy by at most 0.011, and cuts the loss to electrode removal by more than half.

variant loses more than 0.024 BA to noise or jitter. For this architecture, accuracy under electrode loss is thus set mainly by the training procedure (Fig. 4b).

D. External Replication on an Independent Affective Dataset

On DEAP [28] (32 subjects, band-power features, binary valence, within-subject 5-fold), the same pattern appears and is larger (Fig. 4a): at 30% dropout the same features score BA 0.540 under zero-fill (95% CI [0.528, 0.552]) but 0.621 [0.591, 0.648] under mean and 0.638 [0.607, 0.668] under k NN fill, with CIs from resampling the 32 subjects' mean BA. With clean BA 0.639, the fill rule alone moves measured accuracy under dropout by up to 0.11 BA, on a second dataset, task, and montage. This is an external check of the fill-rule effect, not a replication of the full comparison.

VI. DISCUSSION

The main result is methodological: raw retention favors encoders near the chance floor, zero-imputation penalizes non-centered features after standardization, and for trained models augmentation is a third factor. The zero-fill effect agrees with the benchmark finding that channel-dropout fragility depends on whether dropped channels are removed or zero-padded [22], and Proposition 1 gives its mechanism for

standardized features. After accounting for these choices, the fixed reservoir should not be described as the most channel-robust encoder: ERP-window features lead it under every fill at 30% dropout and under every signal-level condition, and zero-fill halves that lead. The reservoir remains useful as a near-centered reference that makes the effect in the other encoders visible, but its invariance belongs to its feature space and does not survive signal-level removal of half the electrodes. Its dynamics still matter: removing the recurrence or driving the reservoir into the irregular regime lowers accuracy, and the pre-specified operating point lies near the measured edge of the ordered regime, in line with edge-of-chaos results for other reservoirs [5], [6].

Learned and generative imputation sit further along the same spectrum (Sec. II). If non-learned fills, constant or k NN, already move measured accuracy by up to 0.13 BA and halve an encoder difference, the fill model of a learned imputer is part of the measured result and should be reported with it. These results suggest a minimal reporting standard: report absolute accuracy with subject-level uncertainty, state and vary the channel-fill rule, and report trained-model results with and without augmentation [24].

Limitations. The primary cohort is a single trial-averaged affective-ERP dataset. The imputation effect nevertheless follows from the standardization algebra of Eq. (5) and is observed across fill rules, dropout rates, and the independent DEAP check, but paradigm-matched multi-cohort ERP validation would strengthen it. EEGNet is trained with one fixed recipe, without a hyperparameter search, and the spectral-radius sweep uses a single draw of the reservoir weights. Adversarial attacks and montage shifts are not evaluated, and graph topology and structure–function coupling are outside the scope of this paper.

VII. CONCLUSION

Channel-dropout robustness rankings for affective-ERP representations depend on the scoring protocol. Raw retention can overstate robustness near the chance floor, zero-imputation shifts non-centered features after standardization and here halves the measured lead of the strongest fixed encoder, and trained models need their augmentation state made explicit. The fixed spiking reservoir, operated near its measured edge of stability, is not the most accurate encoder, but its near-centered code makes these protocol effects visible, and the fill-rule effect replicates on DEAP.

Ethics. The SHAPE study was collected by the Stony Brook University Laboratory for Clinical Affective Neuroscience under that laboratory's Institutional Review Board oversight. DEAP is a publicly released dataset governed by its original collection protocol [28].

Data Availability. For information on the SHAPE study dataset, contact the Laboratory for Clinical Affective Neuroscience at Stony Brook University [29]. DEAP is public [28].

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